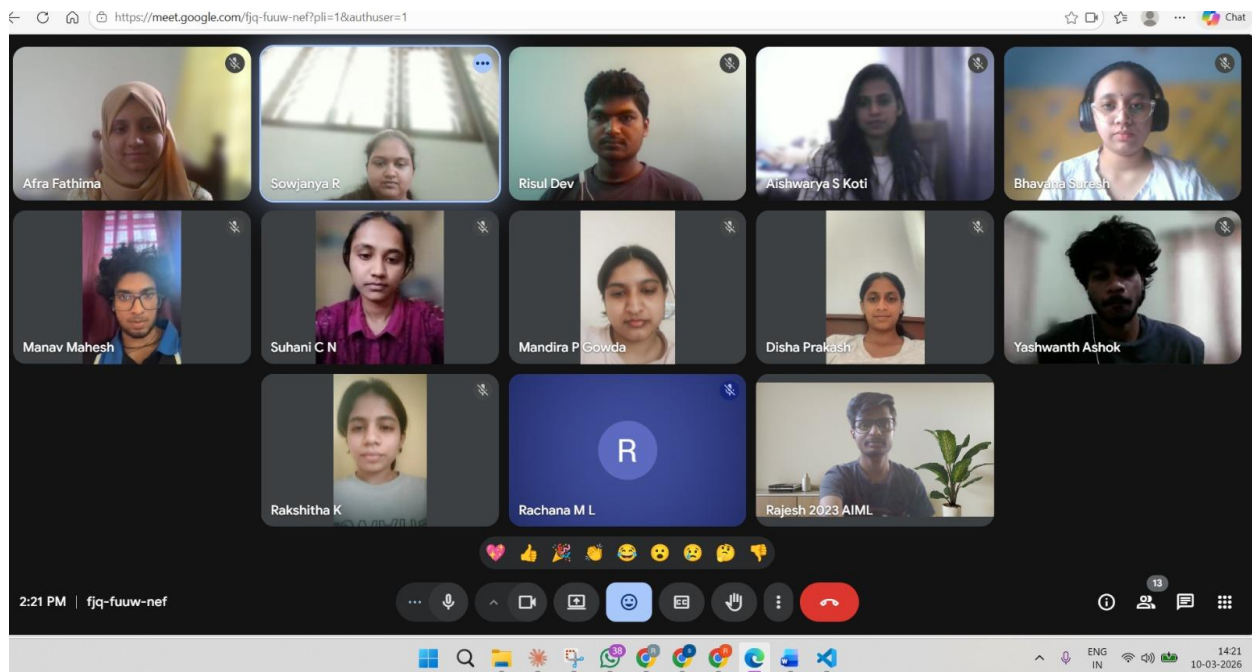


Mini Project: Cohort 21: Forest Fire Ignition Risk

Date:10/03/2026

On Day 4 of the project, the team continued working on the model building phase. In the previous session, the model building team had already started implementing the Decision Tree model using the Algerian Forest Fire Dataset. After completing the initial implementation and understanding how the decision tree works with the dataset, the team decided to proceed with building the second model that had been planned earlier.



For this stage, the team implemented the **Random Forest** algorithm. Random Forest is considered an advanced machine learning technique that works by combining multiple decision trees. Instead of relying on a single tree to make predictions, the algorithm builds several trees and then combines their results to produce the final prediction. Because it uses multiple trees, the model is generally more stable and less likely to overfit the data compared to a single decision tree.

The team implemented the Random Forest model using the dataset and observed how it processes the features in order to classify the forest fire risk levels. Since Random Forest uses multiple decision trees internally, it is capable of capturing more complex patterns within the dataset. This makes it a strong candidate for achieving better prediction performance.

CODE SNIPPETS:

```
mm_scaler = MinMaxScaler()
df_minmax = df.copy()
df_minmax[cols_to_scale] = mm_scaler.fit_transform(df[cols_to_scale])

print("MinMax Scaled - first 5 rows:")
print(df_minmax[cols_to_scale].head())
print()
print("Min values (should be 0):")
print(df_minmax[cols_to_scale].min())
print("Max values (should be 1):")
print(df_minmax[cols_to_scale].max())
```

```
df_scaled.to_csv("forest_fire_scaled.csv", index=False)
print("Saved: forest_fire_scaled.csv")
print("Shape:", df_scaled.shape)
print("Columns:", df_scaled.columns.tolist())
```

Python

— Saved: forest_fire_scaled.csv
Shape: (243, 28)
Columns: ['day', 'month', 'year', 'Temperature', 'RH', 'Ws', 'Rain', 'FFWC', 'DMC', 'DC', 'ISI', 'BUI', 'FWI', 'Classes']

MODEL TRAINING

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
```

Python

Forest_Fire_Risk_RF.ipynb X

C:\> Users > TANZEEM > Downloads > Forest_Fire_Risk_RF.ipynb > import pandas as pd

Generate + Code + Markdown | Run All | Outline ...

Name: Fire_Risk_Label, Length: 194, dtype: int64

```
[36] rf_model = RandomForestClassifier(
      n_estimators=200,
      max_depth=10,
      random_state=42
    )

[37] rf_model.fit(X_train, y_train)

...

[38] y_pred = rf_model.predict(X_test)
      print("Predictions:")
      print(y_pred)
```

... Predictions:
[0 0 0 0 0 1 1 1 0 0 1 1 0 0 2 0 1 1 0 1 1 2 0 1 1 0 0 0 0 1 1 2 2 2 2 2
1 0 2 2 1 0 1 0 0 1 2 0]

```
accuracy = accuracy_score(y_test, y_pred)
print()
print("Accuracy:", accuracy)

print("Confusion Matrix:")
print()
print(confusion_matrix(y_test, y_pred))

print("Classification Report:")
print()
print(classification_report(y_test, y_pred))
```

By the end of the session, the team successfully completed the implementation of the Random Forest model. With this step, the model building phase of the project was completed, as both planned models—Decision Tree and Random Forest—had been developed. The next step in the project will involve the evaluation team analyzing both models and comparing their performance to determine which one produces better results for the dataset.